

Key Findings/Governance Memo

This memo summarizes findings from an explainability audit of a gradient-boosted tree model trained on the COMPAS recidivism dataset. Three methods were applied: SHAP values, LIME, and DiCE counterfactuals. The goal was to assess whether the model's predictions are transparent, fair, and amenable to actionable recourse.

What the Explanations Reveal

SHAP analysis shows that `days_b_screening_arrest`, `decile_score`, and `score_text` are the strongest global predictors. Notably, `race_African-American` carries a small but consistent positive SHAP value across the test set, meaning being Black pushes predicted recidivism risk above average even after controlling for other features. This is consistent with the FPR disparity identified in the diagnostic analysis, where Black defendants faced a false positive rate of 0.367 compared to 0.104 for white defendants. LIME explanations for individual defendants were broadly consistent with SHAP on high-impact features like age and priors, but diverged on lower-ranked features. This divergence is expected given that LIME only guarantees local fidelity, but it means the two methods cannot be used interchangeably for individual-level recourse decisions.

Counterfactual Analysis

DiCE counterfactuals did not require changes to immutable features like race or sex to flip predictions, which is a positive finding. However, the most common recourse path for high-risk defendants was a large increase in age, sometimes to 69 or 70. This is not actionable. A defendant cannot change their age, making these counterfactuals technically valid but practically useless for recourse purposes.

Recommendations

First, constrain future counterfactual generation to genuinely actionable features only, excluding age and decile score. Second, the persistent race signal in SHAP values warrants a full proxy discrimination audit examining whether features like `priors_count` are encoding racial disparities from systemic over-policing. Third, explainability outputs should be reviewed on a rolling basis rather than treated as a one-time compliance exercise.